



CLINICAL PROCEDURES AND GUIDELINES
EXTENSIONS TO STANDING ORDERS

June 2026

Clinical Procedures and Guidelines

As per [HCL22-03 Authority to Practice Policy](#), HEMS Clinical Ltd. (HCL) issues specific Clinical Procedures and Guidelines (CPGs) consistent with the National Ambulance Sector Working Group (NASWG). These are issued in accordance with the Medicines (Standing Orders) Regulations 2002.

A standing order is a written instruction issued by a Medical Practitioner. It authorises a paramedic, who does not have prescribing rights, to administer or supply specified medicines or controlled drugs to patients.

Standing orders require the appropriate training alongside compliance with the relevant legislation by the issuer.

Standing orders are required during aeromedical transport where:

- Timely access to medicines may be required during an emergency.
- The prescriber may be unavailable.
- Patient outcomes may be improved by their use.

The use of standing orders is governed by the legislation within the Medicines (Standing Orders) Regulations 2002 - www.legislation.govt.nz.

A practitioner administering medications under a standing order is accountable for their practice and is obliged to administer the drug in accordance with the standing order, have sufficient knowledge of the medication that they are aware of potential adverse reactions associated with administration, and record the treatment including monitoring following administration.

HCL CPG extensions to national standing orders

These CPG extensions permit clinicians with HCL Authority to Practice (ATP) additional scope of practice to those in the national CPGs.

ATP is the authorisation by the HCL Medical Director to perform clinical procedures and use medicines within a corresponding scope or practice. The applicable scopes for each medicine are listed in this document.

Dosing is for adults only. Where the medicine is appropriate to use with children the paediatric dose is specified separately.

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Fentanyl Infusion

Mechanism of action

- Opiate receptor agonist

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Moderate to severe pain
- ✓ Sedation adjunct for ventilated patient

Contraindications

- ✗ Known severe allergy
- ✗ Respiratory depression without a protected airway

Cautions

- ◇ Peripartum - may cause neonatal respiratory depression
- ◇ Elderly/frail - may have increased or prolonged effect
- ◇ Signs of shock
- ◇ High risk of respiratory depression - COPD/morbid obesity

Dosage

Adult

- Infusion rate: suggested start rate 100mcg/hr or 10mL/hr up to 240mcg/hr or 24mL/hr of 10mcg/mL

Paediatric

- Infusion rate: 2-4mcg/kg/hr or 0.2-0.4mL/kg/hr of 10mcg/mL

Preparation for IV infusion

- Each ampoule contains 100mcg/2mL
- Draw up 200mcg (4mL) in a 20mL syringe and **dilute to 20mL** with NaCl 0.9%
 - This syringe contains: **10 mcg/mL Fentanyl**

Common adverse effects

- Respiratory depression
- Hypotension
- Brady-dysrhythmias
- Sedation
- Nausea and vomiting

Fentanyl / Ketamine Infusion

	Fentanyl	Ketamine
Mechanism of action	Opiate receptor agonist	Predominately NMDA receptor antagonist, inhibition of excitatory neurotransmitters in the brain. Low doses cause analgesia, larger doses cause amnesia and dissociation, and high doses cause anaesthesia
Delegated scope of practice	ICP or CCP	
Indications	✓ Post intubation sedation	
Contraindications	✗ Known severe allergy	
Cautions	⚠ Peripartum - may cause neonatal respiratory depression ⚠ Elderly/frail - increased or prolonged effect ⚠ Signs of shock	

Dosage

Adult:

- Suggested start rate 8mL/hr (0.2mL/kg/hr) and assess effect
- Infusion rate up to 20mL/hr
- Bolus: 2mL PRN (20mcg fentanyl + 20mg ketamine)

Paediatric:

- Suggested start rate 0.2mL/kg/hr (fentanyl 2mcg/kg/hr + ketamine 2mg/kg/hr) and assess effect
- Infusion rate: up to 0.5mL/kg/hr

Preparation for IV infusion

- Each Fentanyl ampoule contains 100mcg/2mL
- Each Ketamine ampoule contains 200mg/2mL
- Draw up 200mcg fentanyl (4mL) and 200mg ketamine (2mL) into a 20mL or 30mL syringe and **dilute to 20mL** with NaCl 0.9%
 - This syringe has: **fentanyl 10mcg/mL + ketamine 10mg/mL**

Common infusion adverse effects

- Bradycardia or Tachycardia
- Hypotension

Fentanyl / Midazolam Infusion

	Fentanyl	Midazolam
Mechanism of action	Opiate receptor agonist	Benzodiazepine – enhanced inhibitory activity of GABA within the central nervous system, resulting in anticonvulsant activity, sedation, amnesia, anxiolysis
Delegated scope of practice	ICP or CCP	
Indications	✓ Post intubation sedation	
Contraindications	✗ Known severe allergy	
Cautions	◇ Peripartum - may cause neonatal respiratory depression ◇ Elderly/frail - increased or prolonged effect ◇ Signs of shock ◇ Concurrent administration of opiates, ketamine or other sedatives - this will increase/prolong effects	

Dosage

Adult:

- Suggested start rate: 8mL/hr (0.2mL/kg/hr)
- Infusion rate: up to 20mL/hr (fentanyl 0-200mcg/hr + midazolam 0-20mg/hr)

Paediatric:

- Suggested start rate: 0.2mL/kg/hr (fentanyl 2mcg/kg/hr and midazolam 0.2mg/kg/hr)
- Infusion rate: up to 0.5mL/kg/hr

Preparation for IV infusion

- Each Fentanyl ampoule contains 100mcg/2mL
- Each Midazolam ampoule contains 15mg/3mL
- Draw up 300mcg fentanyl (6mL) and 30mg midazolam (6mL) into a 30mL or 50mL syringe and **dilute to 30mL** with NaCl 0.9%
 - This syringe has: **fentanyl 10 mcg/mL and midazolam 1 mg/mL**

Common adverse effects

- Bradycardia
- Hypotension

Ketamine Infusion

Mechanism of action

- Predominately N-methyl-d-aspartate (NMDA) receptor antagonist, inhibiting excitatory neurotransmitters in the brain. Low dose for analgesia, moderate for amnesia/dissociation, and higher dose for anaesthesia.

Delegated scope of practice

- Paramedic, ICP or CCP

Indications

- ✓ Severe pain
- ✓ Sedation (including anaesthesia for ventilation)

Contraindications

- ✗ Known severe allergy
- ✗ Aged less than one year

Cautions

- | | |
|--|---|
| ◇ Active psychosis | ◇ Current myocardial ischaemia |
| ◇ Hypertension | ◇ Concurrent administration of sedatives or midazolam |
| ◇ Clinical conditions that may be made worse by hypertension | ◇ Elderly and/or frail |

Dosage

Adult:

- Loading dose: up to 0.25mg/kg to max 25mg; given over 15 minutes of **10mg/mL ketamine**
- Suggested start rate 60mg (6mL)/hr of **10mg/mL ketamine**; up to 1mg/kg/hr to maximum 100mg (10mL)/hr of **10mg/mL ketamine**

Paediatric:

- Loading dose: up to 0.25mg/kg to max 25mg; given over 15 minutes of **1mg/mL ketamine**
- Suggested start rate 0.5mg/kg/hr (0.5mL/kg/hr) of 1mg/mL ketamine; up to 1mg/kg/hr (1mL/kg/hr) of **1mg/mL ketamine**

Preparation for IV infusion

- Each ampoule contains 200mg in 2mL

Adult:

- Draw up 200mg (2mL) ketamine into a 20mL or 30mL syringe and **dilute to 20mL** with 0.9% NaCl
 - This syringe has: **10mg/mL ketamine**

Paediatric:

- Draw up 100mg (1mL) ketamine and add to a 100mL bag of 0.9% NaCl; draw from this into 20mL or 30mL syringe for infusion
 - This syringe has: **1mg/mL ketamine**

Common adverse effects

- | | |
|--------------------------|-----------------------|
| • Transient hypertension | • Nausea and vomiting |
| • Tachycardia | • Sedation |
| • Apnoea | • Hallucination |

Propofol

Mechanism of action

- A short-acting intravenous anaesthetic agent suitable for the induction and maintenance of sedation for mechanical ventilation. Thought to work on GABA-A receptors in the brain by decreasing the dissociation of GABA, potentiating the inhibitory effects of the neurotransmitter.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Maintenance of sedation for mechanical ventilation during transport

Contraindications

- ✗ Known hypersensitivity to propofol or constituents
- ✗ Allergy to egg/egg products or soy

Cautions

- ◇ Safe for use in pregnancy however, due to utero-placental transfer may result in neonatal CNS and respiratory depression
- ◇ Hypotension/shock

Dosage

Adult:

- Infusion at 2-4 mg/kg/hr (0.2-0.4 mL/kg/hr) to a maximum of 30 mL/hr of **10mg/mL propofol**

Paediatric:

- **Seek clinical advice with 0800 HEMSMD before continuation or commencement**
- Infusion at 2-4 mg/kg/hr (0.2-0.4 mL/kg/hr) to a maximum of 20 mL/hr of **10mg/mL propofol**

Preparation for IV infusion

- Each ampoule contains 10mg/mL (1%)
- Draw up 200mg (20 mL) in a syringe, do not dilute.
 - This syringe has: **10mg/mL propofol**

Common adverse effects

- Transient injection site pain
- Hypotension due to vasodilation
- Respiratory depression

Effect

- Rapid dose-dependent onset of action, ~40 seconds with loss of consciousness
- Duration 3-5 minutes following a bolus (increases with repeated or prolonged administration)
- Rapid termination of effect after a single bolus (or short infusion), predominantly due to redistribution

NOTE: this drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with an external medical facility doctor.

Adrenaline Infusion

Mechanism of action

- Endogenous catecholamine that is a direct acting sympathomimetic at α - and β -adrenergic receptors. At lower doses predominantly β_1 - effects with increased heart rate and cardiac output. At higher doses, α_1 - effects manifest with peripheral vasoconstriction to increase SVR, afterload and coronary perfusion.
- β_2 - affects are associated with bronchial smooth muscle relaxation.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Anaphylaxis
- ✓ Severe asthma or Imminent respiratory arrest from COPD
- ✓ Bradycardia causing shock
- ✓ Blood pressure augmentation
- ✓ Septic shock, cardiogenic shock (e.g. post STEMI), neurogenic shock

Contraindications

- ✗ None

Cautions

- ◇ Myocardial ischaemia - adrenaline will increase myocardial oxygen consumption
- ◇ Tachydysrhythmias - adrenaline may make dysrhythmias worse

Dosage

Adult:

- Infusion rate: up to 1.5mg (25mL)/hr to target blood pressure; suggested start rate 180mcg (3mL)/hr (this is the equivalent of 1dp/sec regime of '1mg in 1000mL' adrenaline) of **60mcg/mL adrenaline**

Paediatric:

- infusion rate: 1-10 mcg/kg/hr [0.1-1mL/kg/hr] to target blood pressure of **10mcg/mL adrenaline**

Preparation for IV infusion

- Each ampoule contains 1mg/mL

Adult:

- Draw up 3mg in a 50mL syringe and **dilute to 50mL** with 0.9% NaCl
 - This syringe has: **60mcg/mL adrenaline**

Paediatric:

- Draw up 0.5mg (0.5mL) adrenaline in a 50mL syringe and **dilute to 50mL** with 0.9% NaCl
 - This syringe has: **10mcg/mL adrenaline**

Common adverse effects

- | | |
|--|---|
| <ul style="list-style-type: none">• Tachycardia/Tachydysrhythmia• Myocardial ischaemia• Hypertension• Nausea/vomiting | <ul style="list-style-type: none">• Tremor/anxiety/sweating• Hyperglycaemia• Extravasation can cause digit ischaemia, monitor infusion site carefully |
|--|---|

Noradrenaline Infusion

Mechanism of action

- Endogenous catecholamine with direct α - and β -adrenergic receptor stimulation. Predominantly, α_1 -with peripheral vasoconstriction increases SVR, afterload and coronary perfusion pressure, and venoconstriction increases venous return and preload. Weaker β -activity with inotropy and chronotropy.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Hypotension refractory to fluid resuscitation
- ✓ Primarily distributive shock, e.g. septic shock, neurogenic shock
- ✓ Shock associated with toxic ingestion
- ✓ Blood pressure augmentation, e.g. acute spinal cord injury

Contraindications

- ✗ None

Cautions

- ◇ Myocardial ischaemia - increased afterload will increase myocardial work and oxygen consumption
- ◇ Peripheral ischaemia
- ◇ Potential extravasation and digit ischaemia - monitor infusion site carefully

Dosage

Adult:

- Infusion rate: up to 25mL/hr to target blood pressure; suggested start rate 5mL/hr of **6mcg/mL noradrenaline**

Paediatric:

- Not for paediatric use
- **Seek clinical advice with 0800HEMSMD before continuation or commencement**

Preparation for IV infusion

- Each ampoule contains 1mg/mL
- Draw up 3mg of nor-adrenaline and place in a 500 mL bag of 0.9% NaCl; this is suitable for peripheral administration.
 - This bag has: **6 mcg/mL noradrenaline**
 - Then draw off 50 mL in a 50 mL syringe

NOTE that an alternative concentration of 60mcg/mL may be used if the patient has central line IV access; phone 0800HEMSMD to consult before initiating.

Common adverse effects

- Reflex bradycardia, hypertension
- Myocardial, mesenteric, or peripheral (digit) ischaemia

NOTES:

If extravasation occurs, do not flush the cannula; leave the IV cannula in situ, as the receiving hospital may elect to inject phentolamine. Write on the cannula dressing 'do not use' and occlude the bung. Document clearly on the acute care summary which medication, volume, and time of extravasation.

This drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with a doctor from an external medical facility.

Metaraminol Infusion

Mechanism of action

- α_1 -adrenoceptor agonist. Mechanism of action through both direct α_1 receptor and indirect effects secondary to noradrenaline release. Effect on peripheral blood vessels causing vasoconstriction and associated increase in blood pressure

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Hypotension in the setting of septic shock, post cardiac arrest, cardiogenic shock, severe traumatic brain injury, neurogenic shock, RSI and post intubation

Contraindications

- ✗ Known severe allergy or known allergy to sulphites

Cautions

- ◇ Bradycardia; consider adrenaline infusion for hypotensive, bradycardic patient

Dosage

Adult:

- Infusion rate: up to 20mL/hr; suggested start rate 2mL/hr (2mg/hr) of **metaraminol 1mg/mL**

Paediatric:

- **Seek clinical advice with 0800HEMSMD before continuation or commencement**
- Infusion rate; Infusion rate: 0.1-0.2mL/kg/hr of **metaraminol 0.1mg/mL**

Preparation for IV infusion

- Each ampoule contains 10mg/1mL metaraminol

Adult:

- Draw up 10mg (1mL) in a 10mL syringe and **dilute to 10mL** with 0.9% NaCl.
 - This syringe has: **metaraminol 1mg /mL**

Paediatric:

- Add metaraminol 10mg (1mL) to a 100mL bag of NaCl 0.9%. Draw 10mL from 100mL bag into **10mL syringe**
 - This syringe has: **metaraminol 0.1mg/mL**

Common adverse effects

- Hypertension - usually only occurs when boluses greater than 1mg are administered in an adult
- Bradycardia

Phenylephrine Infusion

Mechanism of action

- Synthetic α_1 -adrenoceptor agonist. Direct α_1 mechanism of action, with effect on peripheral blood vessels causing vasoconstriction and associated increase in blood pressure.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Hypotension in the setting of septic shock, post cardiac arrest, cardiogenic shock, severe traumatic brain injury, neurogenic shock, RSI and post intubation

Contraindications

- ✗ Known severe allergy or known allergy to sulphites

Cautions

- ◇ Bradycardia; consider adrenaline infusion for hypotensive, bradycardic patient

Dosage

Adult:

- Infusion rate: up to 20mL/hr; suggested start rate 2mL/hr (200mcg/hr) of phenylephrine 100mcg/mL

Paediatric:

- Not for paediatric use
- Seek clinical advice with 0800HEMSMD before continuation or commencement

Preparation for IV infusion

- Each ampoule contains 10mg/1mL phenylephrine
- Adult: Draw up 10mg (1mL) into a bag of 100mL 0.9% NaCl. Draw 50mL of this into a 50mL syringe
 - This syringe has: **phenylephrine 100mcg/mL**

Common adverse effects

- Hypertension - usually only occurs when boluses greater than 1mg are administered in an adult
- Bradycardia

NOTE: this drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with an external medical facility doctor.

Isoprenaline Infusion

Mechanism of action

- Isoprenaline hydrochloride is a synthetic sympathomimetic amine that is structurally related to adrenaline but acts almost exclusively on β -adrenoceptors; it is a potent non-selective β -agonist that increases cardiac output by increasing heart rate, contractility and venous return.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Heart block that requires support prior to pacemaker therapy

Contraindications

- ✗ Patients with tachyarrhythmias or tachycardia
- ✗ AV block caused by digitalis toxicity; may aggravate cardiac irritability
- ✗ History of hypersensitivity to isoprenaline

Cautions

- ◇ Recent myocardial ischaemia or cardiogenic shock

Dosage

Adult:

- Infusion rate: start at 7mL/hr (28mcg/hr); up to 25mL/hr to target heart rate (usually HR > 60) of **4mcg/mL of isoprenaline**

Paediatric:

- Not for paediatric use
- **Seek clinical advice with 0800HEMSMD before continuation or commencement**

Preparation for IV infusion

- If an infusion is already running consider **consultation with 0800HEMSMD** around whether to draw up new using HCL preparation (below) or transport on current concentration
- Each ampoule contains 1mg/5mL (200mcg/mL) isoprenaline hydrochloride
- Draw up 2 x ampoules (2mg or 10mL) into 490mL 5% Dextrose (D5W)
- Draw off 50 mL into a 50mL syringe
 - This syringe has: **4 mcg/mL Isoprenaline**

Common adverse effects

- | | |
|---------------|-------------|
| • Hypotension | • Dizziness |
| • Arrhythmias | • Anxiety |
| • Headache | |

Effect

- Rapid dose-dependent onset, transient duration of effect (minutes)

NOTE: this drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with an external medical facility doctor.

Glyceryl Trinitrate Infusion (GTN)

Mechanism of action

- Vascular smooth muscle relaxation with resultant (predominantly venous) vasodilation; reduced myocardial oxygen demand through both pre-load reduction (venous return) and afterload reduction (systemic arterial vasodilation)

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Symptomatic myocardial ischaemia
- ✓ Control of hypertension (in conjunction with labetalol) e.g. autonomic dysreflexia, STEMI, intracranial hypertension

Contraindications

- ✗ Systolic <100
- ✗ HR <40/min or >150/min
- ✗ Ventricular tachycardia

Cautions

- ◇ STEMI where right ventricle may be involved (preload reduction may result in significant reduction in cardiac output); consider consultation with 0800HEMSMD on-call
- ◇ Aortic stenosis or mitral stenosis (inability to compensate for preload reduction)

Dosage

Adult:

- Infusion rate: up to 10mL/hr (10mg/hr) titrated to *maximum* systolic blood pressure; suggested start rate 2mL/hr (2mg/hr) of **1mg/mL of GTN**
 - Adjust the rate as required by 0.5mg/hr every 5 - 10 minutes titrating to effect
 - Bolus: not endorsed

Paediatric:

- Not for paediatric use
- Seek clinical advice with 0800HEMSMD before continuation or commencement

Preparation for IV infusion

- Each ampoule contains 50mg in 10mL
- Draw up 50mg into a 50mL syringe and **dilute to 50mL** with 0.9% NaCl.
 - This syringe has **1mg/mL GTN**

Common adverse effects

- Reduced cardiac output

Labetalol Infusion

Mechanism of action

- Adrenoceptor antagonist, with B₁ effect resulting in reduced heart rate, cardiac output and blood pressure and α₁-effect resulting in vasodilation and associated blood pressure reduction

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Control of hypertension where rapid reduction in blood pressure indicated, e.g. pregnancy induced hypertension, STEMI, acute stroke, unsecured aneurysmal subarachnoid haemorrhage, and suspected or confirmed acute aortic aneurysm/dissection.
- ✓ Control of hypertension associated with autonomic dysreflexia

Contraindications

- ✗ Bradycardia
- ✗ Hypotension
- ✗ Symptomatic cardiac failure
- ✗ Acute severe asthma

Cautions

- ◇ Patients with pre-existing heart failure (may reduce cardiac output)
- ◇ 1st degree heart block, known sick sinus, or previous 2nd / 3rd degree heart block without a pacemaker in situ (may worsen block or cause bradycardia)
- ◇ Asthma/COPD that requires regular bronchodilators (may cause bronchospasm)

Dosage

Adult

- Infusion rate: up to 40mL/hr (200mg/hr) **titrated to maximum systolic blood pressure** as directed by receiving team or HEMS Clinical on-call; suggested start rate at 12mL/hr (60mg/hr) of 5mg/mL of labetalol and assess effect
 - Adjust the rate as required by 4mL/hr (20 mg/hr) every 10 minutes, titrating to effect
- Bolus: 2-4mL (10-20mg) IV as determined by response

Paediatric:

- Seek clinical advice with 0800HEMSMD before continuation or commencement

Preparation for IV infusion

- Each ampoule contains 100mg in 20mL
- Draw up undiluted into 20mL or 30mL syringe
 - This syringe has **5mg/mL labetalol**

Common adverse effects

- Hypotension
- Bronchospasm

Hydralazine

Mechanism of action

- Peripheral vasodilation through direct relaxation of smooth muscle in vascular resistance vessels (predominantly arterioles); the cellular mechanism for this is not well understood.

Preparation

- Hydralazine hydrochloride 20mg/mL powder for injection

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Hypertensive crisis
- ✓ Pregnancy associated hypertension

Contraindications

- ✗ Known hypersensitivity to hydralazine
- ✗ Severe tachycardia or heart failure with high cardiac output e.g. thyrotoxicosis
- ✗ Myocardial insufficiency due to mechanical obstruction e.g. aortic or mitral stenosis, constrictive pericarditis
- ✗ Isolated right heart failure due to pulmonary hypertension (cor pulmonale)
- ✗ Dissecting aortic aneurysm

Cautions

- ◇ Heart failure with hypotension

Dosage

Adult:

- 5-10mg slow IV injection q30 minutes

Paediatric:

- Not for paediatric use
- **Seek clinical advice with 0800HEMSMD before continuation or commencement**

Administration

- Ampoule contains 20mg/mL powder for injection
- Reconstitute with 1mL sterile water for injection; this makes a 20mg/mL solution
 - **Dilute to 20mL** with 0.9% NaCl; this makes a **1mg/mL solution**; mix thoroughly
 - Alternatively, where there is no access to water for injection, reconstitute with 2mL 0.9% NaCl solution and dilute with 18mL of 0.9% NaCl; this makes a **1mg/mL solution**; mix thoroughly
- Give 5-10mL (5-10mg) per dose

Common adverse effects

- Tachycardia, palpitation, anginal symptoms, flushing
- Headache
- Dizziness

Effect

- Onset IV dose ~5 minutes, maximum effect 10-80min, duration 1-4 hours

Amiodarone Infusion

Mechanism of action

- Antidysrhythmic with broad spectrum of activity, predominantly class III action. Prolongs action potential duration, reduces automaticity, prolongs refractory period, reduces abnormal electrical activity and heart rate.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Adults who are not in cardiac arrest but with sustained VT
- ✓ Adults with moderate cardiovascular compromise as a result of rapid atrial fibrillation or rapid atrial flutter

Contraindications

- ✗ Known severe allergy to amiodarone or to iodine
- ✗ Evidence or history of thyroid dysfunction
- ✗ Severe atrioventricular conduction disorders
- ✗ Atrial fibrillation associated with Wolff-Parkinson-White
- ✗ VT secondary to tricyclic antidepressant (TCA) poisoning or cocaine

Cautions

- ◇ Poor perfusion or signs of low cardiac output - amiodarone reduces inotropy and may cause a significant fall in cardiac output, particularly when administered rapidly
- ◇ Hypotension - may cause vasodilation and worsen hypotension, particularly when administered rapidly
- ◇ Atrial fibrillation associated with sepsis - amiodarone may cause a significant fall in cardiac output
- ◇ Known sick sinus syndrome without an internal pacemaker in place - amiodarone slows heart rate and severe bradycardia may occur following reversion of tachydysrhythmia
- ◇ Previous 2nd and 3rd degree heart block without internal pacemaker in place - amiodarone slows heart rate and severe bradycardia may occur following reversion of tachydysrhythmia
- ◇ Pregnancy

Dosage

Adult:

- 300mg IV Infusion; Infusion rate: 300mg over 30 minutes (100ml/hr)
 - A further 150mg IV infusion may be administered over 30 minutes if the tachydysrhythmia persists (50ml/hr)

Paediatric:

- Not for paediatric use
- **Seek clinical advice with 0800HEMSMD before continuation or commencement**

Preparation for IV infusion

- Each ampoule contains 150mg/3mL
- Draw up 300mg amiodarone (6mL) in a 50mL syringe and **dilute to 50mL** with 0.9% NaCl.
 - This syringe has: **6mg/mL amiodarone**
- For additional infusion of the same concentration, draw up 150mg amiodarone (3mL) in a 50mL syringe and dilute to 25mL with 0.9% NaCl.

Common adverse effects

- Hypotension
- Brady-dysrhythmias

Note: to troubleshoot adverse effects the infusion rate can be reduced, or the infusion transiently stopped; advice should be sought through 0800HEMSMD on call.

Promethazine

Mechanism of action

- Antihistaminergic (H₁-receptor block), antimuscarinic (mAChR block) and direct antagonist of mesolimbic (brain) dopaminergic and α-adrenergic receptors.

Preparation

- Promethazine hydrochloride 50mg / 2mL solution for injection

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Allergic reactions, including urticaria, angioedema, and anaphylaxis
- ✓ Relief of upper respiratory tract congestion with allergic rhinitis or hay fever
- ✓ Nausea and vomiting
- ✓ Motion sickness
- ✓ Sedation

Contraindications

- ✗ Known hypersensitivity to promethazine
- ✗ Children under the age of 6 (risk of respiratory depression).

Cautions

- ◇ Epilepsy
- ◇ Risk of tissue damage with extravasation (do not give SC).
 - Caution with intramuscular (IM) - **must be** deep IM if using this route.

Dosage

Adult:

- IV Route Preferred
- 25mg IV may be repeated at 2 hours

Paediatric over 5 years:

- 1 mg/kg IV/IM to a maximum 12.5mg, one dose only

Administration

- Ampoule contains 50mg/2ml
- **Dilute to 10mL** using 0.9% NaCl.
 - This syringe has: **5mg/mL promethazine**
- Then administer the required dose IV over 1 minute to avoid hypotension

Common adverse effects

- Sedation, confusion, and disorientation
- May cause paradoxical excitation, restlessness or agitation and (rarely) seizures
- Blurred vision, dilated pupils, dry mouth (anticholinergic effect)
- Arrhythmia, hypotension (less common)

Effect

- Onset 3-5 minutes, duration up to 12 hours for antihistamine effect (2-8 hours for sedation)

Hydrocortisone

Mechanism of action

- Glucocorticoid, mineralocorticoid.
- Inhibition of the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

Preparation

- Hydrocortisone sodium succinate powder for injection

Delegated scopes of practice

- Paramedic, ICP or CCP

Indications

- ✓ Anaphylaxis
- ✓ Status asthmaticus
- ✓ Urticarial transfusion reactions
- ✓ Acute non-infectious laryngeal oedema e.g. following fibrinolytic therapy for acute occlusive stroke
- ✓ Adrenocortical insufficiency (primary/Addisonian crisis or secondary)
- ✓ Shock unresponsive to conventional therapy if adrenocortical insufficiency known or suspected
- ✓ Sepsis (**Under consultation**)

Contraindications

- ✗ Known hypersensitivity to hydrocortisone or constituent products

Cautions

- ◇ Immunosuppressant effect, increased susceptibility to infection

Dosage

Adult:

- 100mg IV

Paediatric:

- Adrenal Crisis – As per the national EAS CPGs
- Asthma - 4mg/kg IV to max 100mg (or age <5 years give 25-50mg, >5 years give 50-100mg)

Administration

- Press down on the plastic cap to force diluent (2mL sterile water for injection) into the vial; shake until well dissolved; remove the central plastic cap and draw up the contents
 - This solution is **50mg/mL hydrocortisone**
- Administer undiluted IV (or IM)

Common adverse effects

- Fatigue
- Sodium and water retention with repeated dosing

Effect

- Onset 10-20 minutes, duration 8-12 hours

Cyclizine

Mechanism of action

- Cyclizine is a histamine H₁ receptor antagonist (antihistamine) with antimuscarinic properties
- Cyclizine reduces excitability of the inner ear labyrinth and acts directly on the vomiting centre in the brain stem (a central anticholinergic / antimuscarinic action)

Preparation

- 50mg/mL

Delegated scope of practice

- Paramedic, ICP or CCP

Indications

- ✓ Nausea and vomiting associated with motion sickness
- ✓ Nausea and vomiting associated with opiate-based analgesia
- ✓ Vomiting and attacks of vertigo associated with Meniere's disease and other forms of vestibular disturbances

Contraindications

- ✗ Known anaphylaxis and/or adverse drug reaction
- ✗ Acute alcohol intoxication (may increase alcohol toxicity)
- ✗ Porphyria

Cautions

- ◇ Hypotension
- ◇ Severe heart failure or acute myocardial ischaemia (may reduce cardiac output further)
- ◇ Epilepsy, glaucoma, prostatic hypertrophy (or urinary retention)
- ◇ Gastrointestinal obstruction
- ◇ Co-administration of CNS depressants and systemic anticholinergic medications (e.g. atropine)

Dosage

Adult:

- 25mg IV/IM (slow, undiluted), may be repeated once to total of 50mg IV/IM if required

Paediatric:

- 1 - 5 years: 0.5-1mg/kg (to maximum dose 25mg)
- 6 - 16 years: 0.5-1mg/kg (to maximum dose 50mg)

Administration

- Draw up ampoule, concentration will be **50mg/mL**
- Do not dilute

Common adverse effects

- Disorientation, agitation
- Extrapyrarnidal motor disturbances, dyskinesia, dystonia
- Hypotension
- Bronchospasm
- Twitching, muscle spasms

Effect

- Onset 20-30 minutes
- Duration 4 hours

Atropine

Mechanism of action

- Atropine is an anticholinergic agent that acts by competitive antagonism of the action of acetylcholine on muscarinic receptors, causing vagal inhibition. This results in increased heart rate, bronchodilation, drying of respiratory secretions and reduced gastric motility.

Delegated scope of practice

- Paramedic, ICP or CCP

Indications

- ✓ Adults with sinus bradycardia, nodal bradycardia, 1st-degree heart block, 2nd-degree heart block or an undifferentiated narrow complex bradycardia causing clinically significant cardiovascular compromise
- ✓ Organophosphate poisoning
- ✓ Bradycardia associated with intubation in paediatric patients

Contraindications

- ✗ Known severe allergy

Cautions

- ◇ Myocardial ischaemia
- ◇ Down syndrome

Dosages

Adult:

- Bolus dose: 600mcg of **600mcg/mL**
 - Escalating doses may be required for organophosphate poisoning

Paediatric:

- Bolus Dose: 20mcg/kg of **100mcg/mL**
 - Dose may be repeated every 10-15 minutes, to maximum total dose 40mcg/kg
 - Consider consultation with 0800HEMSMD

Preparation for IV Bolus

- Each ampoule contains 600 mcg in 1mL

Adult:

- Undiluted
 - This syringe has **Atropine 600mcg/mL**;

Paediatric:

- Draw 600mcg in a 10mL syringe and **dilute to 6mL** with 0.9% NaCl.
 - This syringe has: **Atropine 100mcg/mL**

Common adverse effects

- | | |
|--------------------------------|-----------------------------------|
| • Tachycardia | • Confusion |
| • Cardiac arrhythmias | • Dry mucous membranes, dry mouth |
| • Hot dry scarlet flushed skin | • Blurred Vision |
| • Hyperthermia | |

Magnesium Infusion

Mechanism of action

- Reduced bronchial smooth muscle contraction resulting in bronchodilation
- Prevention and treatment of eclamptic seizures

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Bronchospasm secondary to severe or life-threatening asthma
- ✓ Severe pre-eclampsia
- ✓ Eclamptic seizure

Contraindications

- ✗ Known severe allergy

Cautions

- ◇ Hypotension (magnesium-associated vasodilation may make hypotension worse)
- ◇ Not to be administered through the same line as sodium bicarbonate or calcium chloride

Dosage

Adult:

- Obstetric (pre-eclampsia/eclampsia): **Load** 20 mmol (5g) over 15 minutes (rate of 160mL/hr); then infusion of 5 mmol/hr (1g/hr) (rate of 10mL/hr)
- Adult Asthma/COPD: **Load** 10mmol over 15 minutes; infusion rate: 80mL/hr (20mL over 15mins)

Paediatric:

- Asthma: 0.1mL/kg of magnesium (0.2mmol/kg). Infusion rate: 80mL/hr (20mL over 15mins)

Preparation for IV infusion

- Each 5 mL ampoule of Magnesium sulphate 50% contains 10 mmol (2.47g) of magnesium

Obstetric (pre-eclampsia/eclampsia): to deliver a 20mmol (5g) load then 1g/hr maintenance.

- Draw up 20mmol (5g) of magnesium sulphate (10mL) into a 50mL syringe
- **Dilute to 40mL** using 0.9% NaCl.
 - This syringe has: **Magnesium 0.5 mmol/mL**

Adult asthma/COPD:

- Draw up 10mmol of magnesium sulphate (5mL) in a 20mL or 30mL syringe
- **Dilute to 20mL** using 0.9% NaCl.
 - This syringe has: **Magnesium 0.5 mmol/mL**

Paediatric asthma:

- Draw up 0.1mL/kg of magnesium (0.2mmol/kg)
- **Dilute to 20 mL** with 0.9% NaCl.
 - This syringe has: **Magnesium 0.01 mmol/kg/mL**

Common adverse effects

- Hypotension
- Muscle weakness (usually only seen with doses > 20 mmol)

NOTE: this medication is in the National Ambulance Sector CPGs. This standing order allows for additional indications for medicine administration.

Calcium Chloride

Mechanism of action

- Calcium is essential for physiological systems including cell membrane function, enzyme reactions, neuromuscular signalling and activity, cardiac electrophysiology, contraction of cardiac and skeletal muscle, and coagulation.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Release syndrome following crush injury
- ✓ Cardiac arrest secondary to hyperkalaemia
- ✓ To antagonise the cardiotoxicity of hyperkalaemia (known or suspected based on ECG changes)
- ✓ Hypovolaemic shock secondary to trauma (**on-consultation**)
- ✓ Haemorrhage requiring transfusion of blood products (**on consultation**)
- ✓ Selected toxicological syndromes - calcium channel blocker ingestion

Contraindications

- ✗ Known severe allergy
- ✗ Known digoxin or digitalis toxicity
- ✗ Patients in ventricular fibrillation, unless suspected or known hyperkalaemia as cause of arrest

Cautions

- Do not administer intramuscular as can cause myonecrosis

Dosage

Adult:

- Trauma with hypovolaemic shock: Administer 6.8mmol (1g) (10mL) alongside significant volume resuscitation
- Calcium channel blocker overdose: Administer 13.6mmol (2g) (20mL) over 5-10min;
 - Dose may be repeated every 20 minutes if cardiovascular instability persists
- Release syndrome following crush injury in adults: Administer 6.8mmol (1g) (10mL)
 - Repeat the dose if signs of hyperkalaemia persist or recur
- Hyperkalaemia with severe ECG changes: Administer 6.8mmol (1g) (10mL)
 - Repeat the dose after 20 minutes if signs of severe ECG changes persist or recur
- Cardiac Arrest secondary to Hyperkalaemia: 6.8 mmol (1g) of calcium chloride IV as a bolus for an adult
 - Repeat the dose after 10 minutes if ROSC is not achieved

Paediatric:

- Seek clinical advice with 0800HEMSMD before continuation or commencement

Preparation for IV infusion

- Each 10mL ampoule of Calcium Chloride contains 6.8mmol (1g) of calcium
- Administer intravenous into a large vein
- Do not admix with any other medicines other than 0.9% NaCl, as precipitation may occur
- Do not administer through a line with packed red blood cells running.

Common adverse effects

- Venous irritation
- Hypotension or transient hypertension
- Flushing

NOTE: this medication is in the National Ambulance Sector CPGs. This standing order allows for additional indications for medicine administration.

Heparin Infusion

Mechanism of action

- Anticoagulant; inhibits clot formation by enhancing the rate at which antithrombin-III neutralises thrombin and factor Xa, preventing conversion of fibrinogen to fibrin

Delegated scope of practice

- ICP or CCP

Indications

- ✓ STEMI, in combination with fibrinolytic therapy
- ✓ Vascular patients with critical limb ischaemia

Contraindications

- ✗ Known severe allergy
- ✗ Recent neurosurgery

Cautions

- ◇ Age over 75 - increased risk intracerebral haemorrhage
- ◇ Clinically significant bleeding e.g. GI tract or epistaxis
- ◇ Pregnancy (note may be administered if breastfeeding)

Dosage

Adult:

- Infusion rate: as per local hospital guidelines, usually 18 U/kg/hr
- **DO NOT** exceed 1800 U/hour or 18mL/hour
- A loading dose (bolus) may be required if you are initiating this treatment for a facility.
 - **Seek clinical advice with 0800HEMSMD for a loading bolus if required.**

Paediatric/neonatal:

- **Seek clinical advice with 0800HEMSMD before continuation or commencement**

Preparation for IV infusion

- Each ampoule contains 5000IU

Adult:

- Draw up 5000IU (1mL) and **dilute to 50mL** with 0.9% NaCl
 - This syringe has **heparin 100 U/mL**

Paediatric/neonatal:

- **Seek clinical advice with 0800HEMSMD before continuation or commencement**

Common adverse effects

- Increased bleeding

NOTE: this drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with an external medical facility doctor.

Hypertonic Saline 3% (HTS)

Mechanism of action

- Hyperosmolar therapies reduce elevated intracranial pressure through two mechanisms:
 - Plasma volume expansion, which reduces plasma haematocrit, viscosity and therefore cerebral blood volume
 - Creation of an osmotic gradient that draws cerebral oedema fluid from brain.

Preparation

- HTS 3% contains 513mmol/L NaCl and has an osmolarity of 1026mOsmol/L

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Serious traumatic brain injury (sTBI) with a GCS ≤ 8 and one or more of the following clinical signs of raised intracranial pressure:
 - Fixed dilated pupil(s)
 - Unilateral neurological signs
 - Deterioration in GCS more than 2 points since the initial assessment

Contraindications

- ✗ None

Cautions

- ◇ Pre-existing severe hyponatremia (risk of central pontine myelinolysis with rapid correction)
- ◇ Decompensated cardiac failure - risk associated with intravascular volume expansion

Dosage

Adult:

- Infuse the 250mL bag (new volume 276mL) at 3 drops/second (over ~30 minutes)

Paediatric:

- 5mL/kg infusion over 30min (not to exceed 250mL)
- Draw the volume required into syringe(s) and administer over 30 minutes

Preparation for Intravascular Infusion

- Each 20mL ampoule of sodium chloride 23.4% contains 4680mg of sodium chloride
- To make sodium chloride 3%, add 26 mL of 23.4% sodium chloride into a 250 mL bag of 0.9% sodium chloride
 - The volume is now 276mL, concentration **3%, osmolarity 1026mOsmol/L**

Administration

- May be given IO or IV
- A dedicated line must be used for HTS
 - **Do NOT infuse through same line as blood or blood products**

Common adverse effects

- Thrombophlebitis, tissue necrosis if extravasated
- Volume overload
- Electrolyte abnormalities e.g. hypernatremia, hyperchloraemic metabolic acidosis

Effect

- Onset 10 minutes, duration 60 minutes

Insulin Infusion

Mechanism of action

- Binds to cell membrane receptors to trigger intracellular messaging that regulates glucose metabolism

Delegated scope of practice

- Paramedic, ICP or CCP

Indications

- ✓ Treatment of diabetes mellitus – for acute management of diabetic ketoacidosis (DKA) or hyperglycaemic hyperosmolar state (HHS) following acute phase fluid resuscitation
- ✓ Hyperinsulinaemic euglycaemic (HIE) therapy for treatment of overdose

Contraindications

- ✗ Known hypersensitivity to insulin
- ✗ Hypoglycaemia

Cautions

- ◇ Hyper- or hypo-glycaemia may occur as a result of inadequate dosing

Dosage

Adult:

- Infusion rate for DKA/HHS as per local guidelines, usually start at 0.1 IU/kg/hr to target reduction in BSL of no more than 3mmol/L/hr

Paediatric:

- Seek clinical advice with 0800HEMSMD before continuation or commencement

For HIE:

- Administration under 0800HEMSMD consultation only:
 - IU/kg bolus then IU/kg/hour infusion
 - **Must** also bolus 50mL 50% dextrose then have concurrent infusion 50mL/hr 50% dextrose

Preparation for IV infusion

- Each ampoule contains 100 IU/mL
- Adult: Draw up 50U (0.5mL) and **dilute to 50mL** with 0.9% NaCl
 - This syringe has **insulin 1 IU/mL**

Common adverse effects

- Hypoglycaemia

Effect

- Onset following bolus of 10-15 minutes, Duration 15-30 minutes

NOTE: this drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with an external medical facility doctor

Oxytocin Infusion

Mechanism of action

- Oxytocin is a uterotonic, it stimulates uterine contractions by causing G-protein coupled receptors to induce a rise in intracellular calcium and contraction in uterine myofibrils. Contraction of myometrium is associated with constriction of spiral arteries and a reduction in uterine blood flow.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Active management of third stage of labour (following birth)
- ✓ Postpartum haemorrhage

Contraindications

- ✗ Known severe allergy

Cautions

- ◇ None

Dosage

Adult:

- Oxytocin is infused at 10IU per hour IV.

Administration

- Ampoule contains 10IU in 1ml.
- Draw 40 IU oxytocin into 50mL syringe, dilute to 40mL with 0.9% NaCl.
 - This syringe has: **1IU/mL of oxytocin**
- Infusion Rate: 10IU/hr (10mL/hr.) of **1IU/mL oxytocin**

Common adverse effects

- Abdominal cramping
- Tachycardia
- Flushing
- Hypotension

Syntometrine

Mechanism of action

- Syntometrine is a combination of two pharmacotherapeutic agents: ergometrine and synthetic oxytocin. The combination results in prolonged therapeutic effect greater than individual uterotonics.
- Oxytocin, see separate section.
- Ergometrine is an ergot alkaloid which stimulates uterine and vascular smooth muscle contraction. Ergometrine increases the amplitude and frequency of uterine contractions and uterine tone, impeding uterine blood flow. Ergometrine produces predominantly vasoconstriction of capacitance vessels. Arterial vasoconstriction also occurs, through stimulation of alpha-adrenergic and serotonin receptors and inhibition of endothelial-derived relaxation factor release.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Ongoing significant postpartum haemorrhage (PPH) despite first line uterotonic agents
- ✓ PPH with high risk factors

Contraindications

- ✗ Known severe allergy or hypersensitivity to other ergot alkaloids
- ✗ Known latex allergy
- ✗ Essential hypertension
- ✗ Pregnancy associated hypertension
- ✗ Pre-eclampsia or eclampsia
- ✗ Suspected retained placenta
- ✗ Sepsis
- ✗ Peripheral vascular disease or heart disease

Cautions

- ◇ Bradycardia

Dosage

Adult:

- Syntometrine is given as a 1mL IM dose

Administration

- Each 1mL vial contains oxytocin 5IU and ergometrine 500mcg
- Administer undiluted as **intramuscular** injection into lateral thigh
- Repeat dose once, at least 2 hours after first dose
- ***Consider co-administration of anti-emetic medication due to associated nausea/vomiting***

Common adverse effects

- Headache
- Abdominal pain
- Hypertension (may enhance pressor effect of vasoconstricting drugs)
- Bradycardia
- Nausea
- Vomiting
- Chest pain (transient)

Carboprost

Mechanism of action

- Carboprost is a methylated analogue of naturally occurring prostaglandin F2 that stimulates myometrial contractions and vasoconstriction by increasing the number of oxytocin receptors in the myometrium.

Delegated scope of practice

- ICP or CCP

Indications

- ✓ Treatment of postpartum haemorrhage due to uterine atony that has failed to respond to first line mechanical and pharmaceutical therapies

Contraindications

- ✗ Known severe allergy
- ✗ Must not be used for induction of labour or if there is an undelivered baby
- ✗ Acute pelvic inflammatory disease
- ✗ Known cardiac, pulmonary, renal, or hepatic disease.

Cautions

- ◇ Asthma

Dosage

Adult:

- Prostine 15M 250mcg/mL (1mL) solution for IM injection

Administration

- Ampoule contains 250mcg/mL
- Give 250mcg (1mL) by deep intramuscular injection into lateral thigh
- May be repeated every 15 minutes if bleeding continues
- **DO NOT** give by intravenous route

Common adverse effects

- Hyperthermia
- Flushing
- Nausea
- Vomiting

NOTE: this drug is not included in the HCL Medicine Manifest; this CPG extension allows for continuation of a drug initiated in consultation with an external medical facility doctor.

Cefazolin (< 12 years)

Mechanism of action

- First generation cephalosporin antibiotic with activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

Delegated scope of practice

- Paramedic, ICP or CCP

Indications

- ✓ Sepsis where the source of infection appears to be the soft tissues or joint and one or more clinical features indicating antibiotics are present
- ✓ Compound fractures
- ✓ Large contaminated wound
- ✓ Following chest decompression
- ✓ Following amputation
- ✓ Cellulitis - in this setting a single dose may be administered if the patient is being not being directly referred (or transported) to a medical facility.

Contraindications

- ✗ Known severe allergy to cephalosporins

Cautions

- ◇ None.

Dosage

Paediatric:

- Seek clinical advice with 0800HEMSMD before continuation or commencement
- 20mg/kg for all patients

Weight / Age	Dose
5kg / < 1 year	100mg
10kg / 1 year	200mg
20kg / 2-5 years	400mg
30kg / 6-10 years	600mg
40kg / 11-13 years	800mg

Preparation for IV infusion

- The ampoule contains 1g of cefazolin
- Add approximately 4mL of 0.9% NaCl to a 1g ampoule and shake until dissolved.
- Draw up the ampoule contents and **dilute to 10mL**.
- Discard to the desired dose
- Administer IV over 1-2 minutes into a running line.

Common adverse effects

- Rapid administration can cause pain and phlebitis, which is why it should be administered into a running line

NOTE: this medication is in the National Ambulance Sector CPGs. This standing order allows for additional indications for medicine administration.